

3-assignment-esm-262

AUTHOR

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Air Quality and For Loops

```
# Load all necessary libraries
```

```
library(ggplot2)
```

Pt. 1 Build the dataset

```
day <- 1:365
airquality = data.frame(
  day = day,
  season = ifelse(day <= 90, "winter",
                  ifelse(day <= 180, "spring",
                  ifelse(day <= 270, "summer", "fall" ))),
  pm25 = rnorm(365, mean = 30 , sd = 10),
  risk = "NA"
)
```

Pt. 2 Classify Risk with a For Loop

- "good" — PM2.5 below 12 $\mu\text{g}/\text{m}^3$
- "moderate" — PM2.5 between 12 and 35.4 $\mu\text{g}/\text{m}^3$
- "unhealthy" — PM2.5 above 35.4 $\mu\text{g}/\text{m}^3$

```
airquality$consecutive <- as.numeric(0)

for (i in 1:nrow(airquality)) { # loop each row in df
  if (airquality$pm25[i] > 35.4 ) { # designates pm25 column for i values
    airquality$risk[i] <- "unhealthy" # add [i] to only fill in that cell not ev
  } else if (airquality$pm25[i] < 12.0) {
```

```

    airquality$risk[i] <- "good"
  } else {
    airquality$risk[i] <- "moderate"
  }

  if(airquality$risk[i] == "unhealthy") {
    if (i == 1) {
      airquality$consecutive[i] <- 1
    } else {
      airquality$consecutive[i] <- airquality$consecutive[i-1] +1
    }
  } else {
    airquality$consecutive[i] <- 0
  }
}

max_unhealthy_streak <- max(airquality$consecutive)

sprintf("The maximum unhealthy days with PM2.5 over 35.4 µg/m³ is %d days.", max_

```

```
[1] "The maximum unhealthy days with PM2.5 over 35.4 µg/m³ is 4 days."
```

Pt. 3 Seasonal Summary

```

airquality$seasonal_max <- as.numeric(0) # initialize column

for (season in unique(airquality$season)) { # outer loop, reset per season
  seasonal_max <- 0
  for (i in 1:nrow(airquality)) { # loop each row in df
    if(airquality$season[i] == season) { # outer loop to loop by season
      if (airquality$consecutive[i] > seasonal_max) {
        seasonal_max <- airquality$consecutive[i]
      }
    }
  }
  airquality$seasonal_max[airquality$season==season] <- seasonal_max
}

```

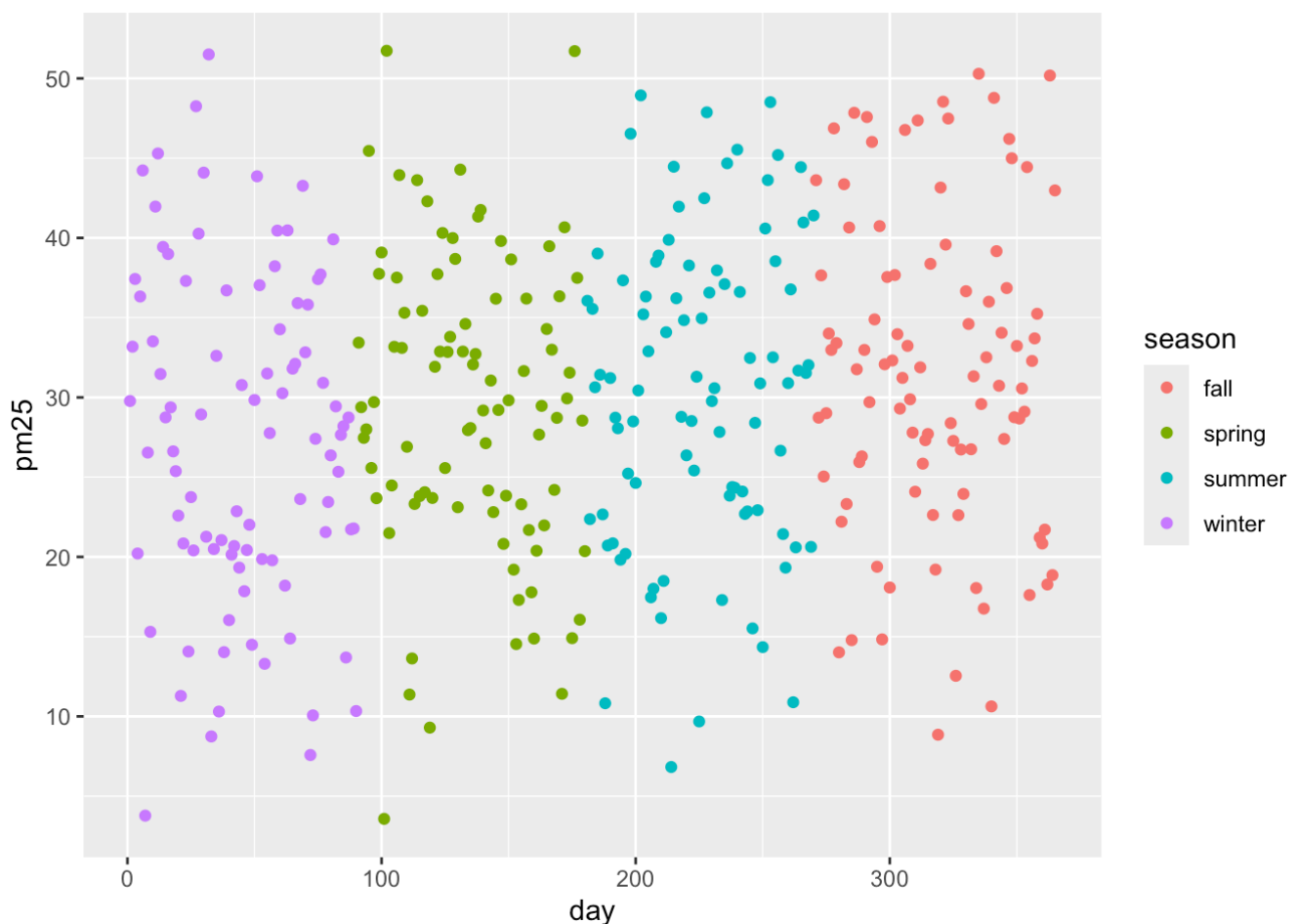
Pt. 4 Visualize and Report

Time Series of PM2.5 across year,

```
annualpm25 <- ggplot(data = airquality, mapping = aes(x = day, y = pm25)) +
  geom_point(aes(color = season))
labs(
  title = sprintf("Daily PM25 2025 Seasonal Variation with
x = \"Day\",
y = \"Particulate Matter 2.5 (µg/m³)\",
) +
  theme_minimal()
```

NULL

annualpm25



- Bar chart of risk category counts (7 points)

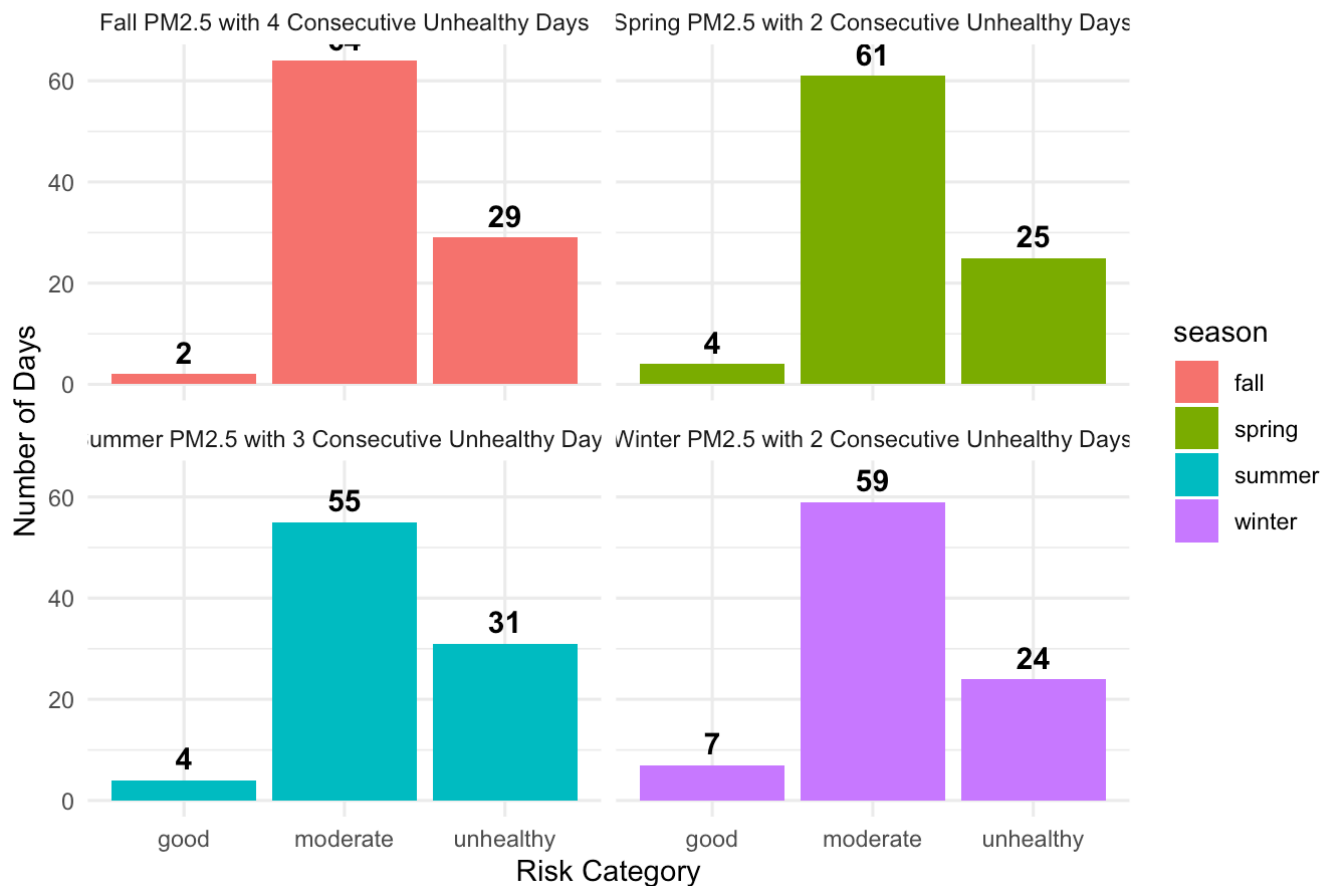
```
# create max per season variable
winter_max <- max(airquality$seasonal_max[airquality$season == "winter"])
spring_max <- max(airquality$seasonal_max[airquality$season == "spring"])
```

```
summer_max <- max(airquality$seasonal_max[airquality$season == "summer"])
fall_max    <- max(airquality$seasonal_max[airquality$season == "fall"])

annualrisk <- ggplot(data = airquality, mapping = aes(x = risk)) +
  geom_bar(aes(fill = season)) +
  geom_text(stat = "count",
            aes(label = after_stat(count)),
            vjust = -0.5,
            size = 4,
            fontface = "bold") +
  facet_wrap(~season,
            labeller = labeller(season = c(
  "winter" = sprintf("Winter PM2.5 with %d Consecutive Unhealthy Days",
    winter_max),
  "spring" = sprintf("Spring PM2.5 with %d Consecutive Unhealthy Days",
    spring_max),
  "summer" = sprintf("Summer PM2.5 with %d Consecutive Unhealthy Days",
    summer_max),
  "fall"    = sprintf("Fall PM2.5 with %d Consecutive Unhealthy Days",
    fall_max)
))) +
  labs(
    title = "Seasonal PM25 Risk Levels",
    x = "Risk Category",
    y = "Number of Days",
  ) +
  theme_minimal()

annualrisk
```

Seasonal PM25 Risk Levels



The annual PM2.5 readings indicate the majority of days across seasons fall in the moderate air quality category, with very few days across all seasons categorized as good air quality days. Spring saw the highest amount of unhealthy days at 67 total days. However, Summer had the most consecutive unhealthy air quality days at 4 in a row. This area is a good venue for increased air quality regulation to increase total good days and reduce the amount of unhealthy days.